

F The Optimality of Oblique Projections

As noted in subsection 4.2, the optimal affine erasure function $r(\mathbf{x}) = \mathbf{b} + \mathbf{P}\mathbf{x}$ does *not* in general use an orthogonal projection for the matrix $\mathbf{P}$. A simple example illustrates why. Let $d = 2, k = 1$ so that X takes values in $\mathbb{R}^2$ and Z takes values in $\mathbb{R}$, with the first feature X_1 and the label Z each independently and uniformly distributed in $\{-1, +1\}$, and the second feature X_2 simply equal to the sum $X_2 = X_1 + Z$. A dataset reflecting such a distribution has four $(\mathbf{x}, y)$ pairs:

$$([1, 2]^T, 1), \quad ([1, 0]^T, -1), \quad ([-1, 0]^T, 1), \quad ([-1, -2]^T, -1)$$

In this case, *all* of the information X has about Z resides in X_2 , so the minimally disruptive orthogonal projection which guards Z will nullify that component:

$$\mathbf{P}_{\text{ortho}} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

On the other hand, X_1 contains some information about X_2 (despite having no information about Z), allowing a partial reconstruction of X_2 while preserving full concept erasure:

$$\mathbf{P}_{\text{oblique}} = \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$$

Both methods fully erase the ability to predict Z from the data, however a simple calculation shows the second, oblique method to perform better as measured by mean squared edit distance:

$$\mathbb{E}\|\mathbf{P}_{\text{ortho}}\mathbf{X} - \mathbf{X}\|^2 = 2, \quad \mathbb{E}\|\mathbf{P}_{\text{oblique}}\mathbf{X} - \mathbf{X}\|^2 = 1$$

G Equivalence of Guardedness Definitions

Xu et al. [43] define the **conditional $\mathcal{V}$ -entropy** of Z given X as the lowest achievable cross-entropy loss predicting Z with a function of X in the predictor class $\mathcal{V}$. In our notation:

$$H_{\mathcal{V}}(Z | X) = \inf_{\theta \in \Theta} \mathbb{E}[\mathcal{L}(\eta(X; \theta), Z)],$$

where $\mathcal{L}(\eta, z) = -\log \frac{\exp(\eta_z)}{\sum_{i=1}^k \exp(\eta_i)}$ is the cross-entropy loss function.

They then define the (unconditional) **$\mathcal{V}$ -entropy** $H_{\mathcal{V}}(Z) = H_{\mathcal{V}}(Z | \mathbf{0})$ to be the lowest achievable cross-entropy loss in the case of a constantly null random data variable. This is exactly our trivially attainable loss L_{τ} (Definition 2.2).

Finally, they define the **$\mathcal{V}$ -information** from X to Z as the reduction in $\mathcal{V}$ -entropy as compared to using such a null random data variable:

$$I_{\mathcal{V}}(X \rightarrow Z) = H_{\mathcal{V}}(Z) - H_{\mathcal{V}}(Z | X).$$

Using these notions, Ravfogel et al. [33] say that X is **ϵ -guarded** with respect to $\mathcal{V}$ if $I_{\mathcal{V}}(X \rightarrow Z) < \epsilon$.

In Appendix B, we showed the equivalence of guardedness (as we have defined it in Definition 2.1) to the optimality of the trivially attainable loss. That is, X ($\mathcal{V}, \mathfrak{L}$)-guards Z when $H_{\mathcal{V}}(Z | X) = L_{\tau} = H_{\mathcal{V}}(Z)$, in the case where $\mathfrak{L}$ is the singleton class consisting solely of the cross-entropy loss function. In the language of [33], X is ϵ -guarded with respect to $\mathcal{V}$ for all $\epsilon > 0$.

H Constraining Norm Growth

In early concept scrubbing experiments (Sec. 6), we found that at specific layers in some models, concept scrubbing with LEACE would cause the norm of the embedding to diverge, leading to NaN outputs. By contrast, SAL never caused divergence, even though it causes a larger disruption to model performance on average (Table 1). This is because SAL uses an orthogonal projection $\mathbf{Q}$, whose eigenvalues are thus all in $\{0, 1\}$, so the norm of the hidden state can never increase after erasure, while LEACE's oblique projection matrix $\mathbf{P}$ does generally have singular values greater than 1. To combine the superior average-case MSE of LEACE with the stability of SAL, we adopt a simple regularization heuristic. After constructing $\mathbf{P}$, we analytically compute the trace of the covariance matrix of the

hidden states after applying $\mathbf{P}$. If $\text{tr}(\mathbf{P}\Sigma_{\mathbf{X}\mathbf{X}}\mathbf{P}^T) > \text{tr}(\Sigma_{\mathbf{X}\mathbf{X}})$, we solve a quadratic equation to find the convex combination $\mathbf{P}' = \alpha\mathbf{P} + (1 - \alpha)\mathbf{Q}$ such that $\text{tr}(\Sigma_{\mathbf{X}\mathbf{X}}) = \text{tr}(\mathbf{P}'\Sigma_{\mathbf{X}\mathbf{X}}(\mathbf{P}')^T)$. By Theorem 4.1, the set of matrices which ensure linear guardedness is convex,¹⁰ so $\mathbf{P}'$ is guaranteed to be in the feasible set. Furthermore, since our mean squared error objective is convex, $\mathbf{P}'$ is guaranteed to have no worse MSE than $\mathbf{Q}$. We find this solves the divergence issue in practice.

I Oracle LEACE

The concept erasure method derived in Section 4 does not require access to concept labels at inference time. That is, we can fit an erasure function on a labeled training dataset, then apply the function to unlabeled datapoints. If we have oracle access to the label z for each x , we can achieve an even more surgical edit. In Theorem I.1 below, we derive **Oracle LEACE**, a closed-form formula for the the nearest X' to any X such that $\text{Cov}(X', Z) = \mathbf{0}$.

Like in Sec. 4, the resulting X'_{LEACE} is “nearest” to X with respect to all p.s.d. inner products $\mathbf{a}^T \mathbf{M} \mathbf{b}$ defined on $\mathbb{R}^d$ simultaneously. This is because, by expressing X in a basis that diagonalizes $\mathbf{M}$, we can decompose the problem into d independent subproblems, one for each component of X_i . Each subproblem can then be viewed as an orthogonal projection, not in $\mathbb{R}^d$, but in an abstract vector space of real-valued random variables. For geometric intuition, see Figure 5.

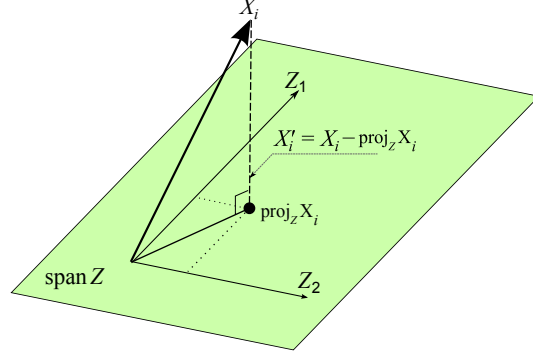


Figure 5: Orthogonal projection of i^{th} component of X , itself a vector in the random variable Hilbert space $\mathcal{H}$, onto the span of the components of Z . The residual $X_i - \text{proj}_Z X_i$ is the closest vector to X_i orthogonal to, and hence uncorrelated with, $Z = \text{span}(\{Z_1, Z_2\})$.

Prior work has noted that computing an orthogonal projection in a random variable Hilbert space is equivalent to solving an ordinary least squares regression problem [1]. Our theorem is a natural extension of this work: we find that X'_{LEACE} is equal to the OLS residual from regressing X on Z , plus a constant shift needed to ensure that erasing Z does not change the mean of X .

Theorem I.1 (Oracle Concept Erasure). *Let $\mathcal{H}$ be the Hilbert space of square-integrable real-valued random variables equipped with the inner product $\langle \xi, \zeta \rangle_{\mathcal{H}} := \mathbb{E}[\xi\zeta]$. Let (X, Z) be random vectors in $\mathcal{H}^d$ and $\mathcal{H}^k$ respectively. Then for every p.s.d. inner product $\langle \mathbf{a}, \mathbf{b} \rangle_{\mathbf{M}} = \mathbf{a}^T \mathbf{M} \mathbf{b}$ on $\mathbb{R}^d$, the objective*

$$\underset{X' \in \mathcal{H}^d}{\text{argmin}} \mathbb{E} \|X' - X\|_{\mathbf{M}}^2 \quad \text{subject to } \text{Cov}(X', Z) = \mathbf{0}$$

is minimized by the (appropriately shifted) ordinary least squares residuals from regressing X on Z :

$$X'_{\text{LEACE}} = X - \Sigma_{\mathbf{X}\mathbf{Z}} \Sigma_{\mathbf{Z}\mathbf{Z}}^+ (Z - \mathbb{E}[Z]).$$

Proof. Assume w.l.o.g. that X and X' are represented in a basis diagonalizing $\mathbf{M}$, so we may write

$$\mathbb{E} \|X' - X\|_{\mathbf{M}}^2 = \sum_{i=1}^d m_i \mathbb{E} [(X'_i - X_i)^2],$$

where $m_1, \dots, m_d \geq 0$ are eigenvalues of $\mathbf{M}$. Crucially, each term in this sum is independent from the others, allowing us to decompose the primal problem into d separate subproblems of the form $\|X'_i - X_i\|_{\mathcal{H}}^2$, one for each component i of (X, X') .

Factoring out constants. Now consider the subspace $\mathcal{C} = \text{span}(1) \subset \mathcal{H}$ consisting of all constant (i.e. zero variance) random variables. Orthogonally decomposing X_i along $\mathcal{C}$ yields $X_i = \tilde{X}_i + \mu_i$, where $\mu_i = \mathbb{E}[X_i] \in \mathcal{C}$ and $\tilde{X}_i = X_i - \mathbb{E}[X_i] \in \mathcal{C}^\perp$, and likewise for X'_i . Our objective is now

$$\|X'_i - X_i\|_{\mathcal{H}}^2 = \|\mu'_i - \mu_i\|_{\mathcal{H}}^2 + \|\tilde{X}'_i - \tilde{X}_i\|_{\mathcal{H}}^2. \quad (7)$$

¹⁰In fact, it is a subspace of $\mathbb{R}^{d \times d}$. For any matrices $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{d \times d}$ such that $\mathbf{A}\Sigma_{\mathbf{X}\mathbf{Z}} = \mathbf{0}$ and $\mathbf{B}\Sigma_{\mathbf{X}\mathbf{Z}} = \mathbf{0}$, we have by linearity $(\alpha\mathbf{A} + \beta\mathbf{B})\Sigma_{\mathbf{X}\mathbf{Z}} = \alpha\mathbf{A}\Sigma_{\mathbf{X}\mathbf{Z}} + \beta\mathbf{B}\Sigma_{\mathbf{X}\mathbf{Z}} = \alpha\mathbf{0} + \beta\mathbf{0} = \mathbf{0}$ for any scalars α and β .

Since μ'_i and μ_i are orthogonal to $\tilde{X}'_i$ and $\tilde{X}_i$, and the constraint $\text{Cov}(X', Z) = \mathbf{0}$ is invariant to constant shifts, we can optimize the two terms in Eq. 7 independently. The first term is trivial: it is minimized when $\mu'_i = \mu_i$, and hence $X'_i = \tilde{X}'_i + \mathbb{E}[X_i]$.

Orthogonal projection. We can now rewrite the zero covariance condition as an orthogonality constraint on $\tilde{X}_i$. Specifically, for every $i \in \{1 \dots d\}$ we have

$$\underset{\tilde{X}'_i \in \mathcal{H}}{\text{argmin}} \|\tilde{X}'_i - \tilde{X}_i\|_{\mathcal{H}}^2 \quad \text{s.t.} \quad \forall j \in \{1 \dots k\} : \langle \tilde{X}'_i, \tilde{Z}_j \rangle_{\mathcal{H}} = 0, \quad (8)$$

where $\tilde{Z} = Z - \mathbb{E}[Z]$. In other words, we seek the nearest $\tilde{X}'_i$ to $\tilde{X}_i$ orthogonal to $\mathcal{Z} = \text{span}(\{\tilde{Z}_1, \dots, \tilde{Z}_k\})$, which is simply the orthogonal projection of $\tilde{X}_i$ onto $\mathcal{Z}^\perp$. This in turn is equal to the ordinary least squares residual from regressing $\tilde{X}$ on $\tilde{Z}$:

$$\tilde{X}'_i = \tilde{X}_i - \text{proj}(\tilde{X}_i, \mathcal{Z}) = X_i - (\Sigma_{XZ})_i \Sigma_{ZZ}^+(Z - \mathbb{E}[Z]) - \mathbb{E}[X_i]. \quad (9)$$

Putting it all together. Plugging Eq. 9 into $X'_i = \tilde{X}'_i + \mathbb{E}[X_i]$ and combining all components into vector form yields

$$X'_{\text{LEACE}} = X - \Sigma_{XZ} \Sigma_{ZZ}^+(Z - \mathbb{E}[Z]), \quad (10)$$

which completes the proof. $\square$

J Notation Key

$\mathcal{Z}$	The space of one-hot labels $\{(z_1, \dots, z_k) \in \{0, 1\}^k \mid \sum_{j=1}^k z_j = 1\}$ (treated interchangeably with the integers $\{1, \dots, k\}$ when convenient).
X, Z	Integrable (i.e. finite first moment) random vectors taking values in $\mathbb{R}^d$ and $\mathbb{R}^k$ respectively (or their realized values inside an expectation, e.g. in $\mathbb{E}[f(X)]$). Z is sometimes restricted to the one-hot labels $\mathcal{Z}$, in which case we assume each $\mathbb{P}(Z = j) > 0$.
X_i, Z_j	The i^{th} and j^{th} components thereof, themselves scalar random variables (or their realized values inside an expectation).
ξ, ζ	Scalar random variables taking values in $\mathbb{R}$.
η	A predictor function $\mathbb{R}^d \rightarrow \mathcal{Z}$ (or its value $\eta(X)$ when inside an expectation).
$\mathcal{V}$	A space of predictor functions $\{\eta(\cdot; \theta) : \mathbb{R}^d \rightarrow \mathbb{R}^k \mid \theta \in \Theta\}$, parameterized by θ and containing all constant functions.
$\mathcal{L}$	A space of loss functions $\{\mathcal{L} : \mathbb{R}^k \times \mathcal{Z} \rightarrow [0, \infty)\}$.
r	An erasure function $\mathbb{R}^d \rightarrow \mathbb{R}^d$, hopefully making a minimal edit to X that eliminates the ability to predict labels Z with predictors in $\mathcal{V}$.
$\mathbf{A}$	A matrix with entries in $\mathbb{R}$.
A_{ij}	The entry thereof at the i^{th} row and j^{th} column.
$\mathbf{A}^+$	The Moore-Penrose pseudoinverse of $\mathbf{A}$.
$\mathbf{v}$	A column vector with entries in $\mathbb{R}$.
v_i	The i^{th} component thereof.